Perceptual Crowdsourcing Under Bias: A Reinforcement Learning Framework for Urban Image Comparisons

Felipe Moreno-Vera and Jorge Poco · Fundação Getúlio Vargas (FGV) · {felipe.moreno, jorge.poco}@fgv.br

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Motivation

Anonymous raters compare pairs of street-level images. Votes become **one consensus score per image**, used downstream as a proxy for objective urban quality.

Aggregation assumes annotators are exchangeable: any two people comparing the same pair would, on average, agree.

That is not obviously safe: annotators differ in **activity**, in a tendency to prefer **one side of the screen**, and possibly in **culturally shaped** notions of what "looks safe".

Dataset: Place Pulse 2.0, Safety

Which one looks safer?



Community of Favela | City Center

Annotators vote on pairs of street-level images.

503,290 pairwise votes

48,358 anonymous users

108,411 street-level images

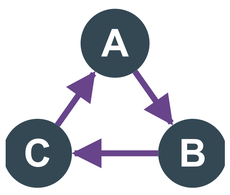
Median user: **7 votes** (mean 10.4, max 2,337).

Each vote records a left image, a right image and an outcome (left, right or equal). Images carry city, country and continent. **Nothing is known about the annotator** beyond an anonymized ID.

0.39% of unique image pairs are judged by more than one user (1,954 of 496,844). This **rules out classical agreement statistics** (Kendall's W, Krippendorff's α).

Diagnosing Annotator Bias

Transitivity violations



A closed triangle of net majorities is **cyclic** if A beats B, B beats C and C beats A.

171	35	20.5%
triangles close	are cyclic	violation rate

Structure beyond random noise, but so few triangles make it **directional, not tightly bounded**. Per annotator it cannot be computed: 0 of 5,749 users with ≥ 20 votes close one.

Consensus residuals

Votes often depart from the consensus (std 0.414), but the per-user mean is unimodal: **no opposing camps. Heavier annotators deviate less,** so bias concentrates among low-activity raters.

−0.242 residual size vs. activity (r)

Effective annotator count

N_{eff} = 28,259 effective annotators 48,358 nominal

58.4%

Moderate concentration: the ten most active users cast only 1.46% of votes.

Annotator Influence Score (AIS)

RMS change in consensus ratings when one user's votes are removed (20 sampled users):



Vote count predicts influence poorly; it appears driven by **how sparsely judged** a user's images are.

Score stability under user resampling

Dropping 20% of users, 20 times: median stability 0.519 (IQR 0.023–0.707). **About 10% of images** swing wildly or reverse sign.

≈10% of images are fragile

Three Ways to Correct It

32-d embedding — Overfits

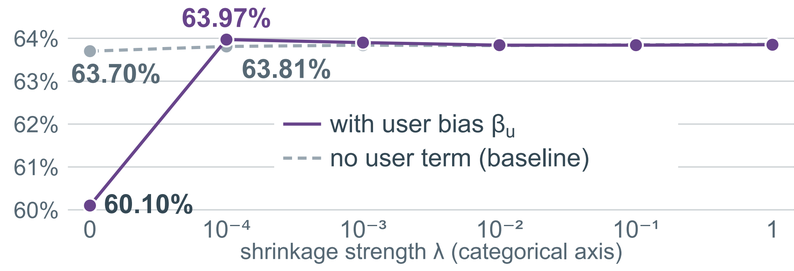
$$\text{score}(u,l,r) = (b_l - b_r) + u_u \cdot (v_l - v_r)$$

63.84% vs. 63.79% without personalization: within noise. With much less weight decay, accuracy falls to **59.50%**.

Positional bias β_u — Generalizes

$$\text{score}(u,l,r) = (s_l - s_r) + \beta_u$$

One left/right lean per user, with a Gaussian prior. At $\lambda = 10^{-4}$: **63.97%** vs. 63.81% without β_u (**+0.16 points**).

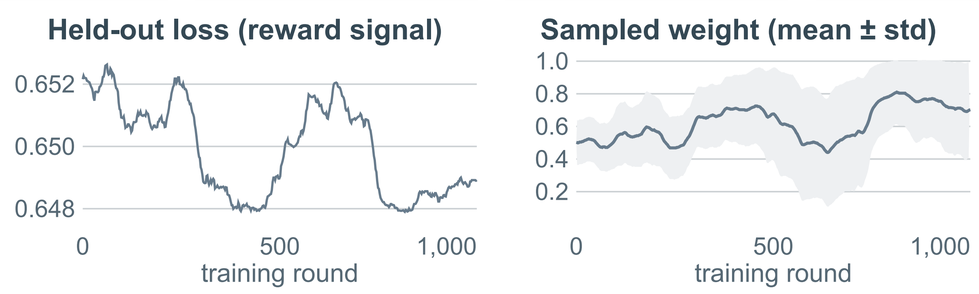


Too little shrinkage overfits; too much collapses onto the baseline. Labeled values are from the paper; the rest are read from its Fig. 1.

RL vote weighting — No gain

Contextual bandit: learns a per-vote trust weight with REINFORCE. The margin over the baseline appears too small relative to batch-level noise.

State s ∈ ℝ⁷ activity, entropy, lean, β_u, residuals ▸ **Policy** Beta(α(s), β(s)), with α, β ≥ 1 ▸ **Refit** Bradley-Terry on sampled weights ▸ **Reward** negative held-out loss



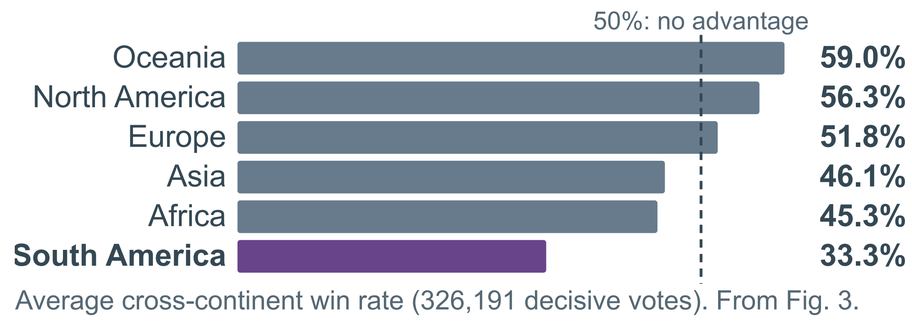
The reward cycles instead of converging; a 300-round rerun ranked the same. Digitized from Fig. 2.

All schemes, same split (best per column in color)

Weighting scheme	Held-out loss	Accuracy
Unweighted (uniform)	**0.6472**	63.88%
β_u shrinkage heuristic	0.6475	**64.02%**
RL policy	0.6484	63.81%
Activity-based weighting	0.6493	63.63%

Bias Beyond the Annotator

Geography of the compared images



Average cross-continent win rate (326,191 decisive votes). From Fig. 3.

- **South America loses to all five other continents,** e.g. 31.8% against North America and 28.1% against Oceania.
- Permutation p ≈ 0 for **14 of 15 pairs** (exception: Africa vs. Asia, p = 0.459), yet continent explains only **6–8%** of total score variance.

Safety vs. wealth

0.355	0.353
Pearson r, unweighted	r after β_u

Over 104,919 images. It is **not proof of bias**: wealth and safety may correlate physically, and this study does not separate the two explanations.

Conclusions & Next Steps

Under realistic sparsity, simple regularized corrections can compete with or beat much more complex learned ones.

- **Parameter count is the binding constraint:** at about 7 votes per annotator, 1 parameter generalizes and 32 do not.
- **Possibly the wrong unit:** annotator weights cannot fix geographic bias.
- **Limits:** safety focus; 20-user AIS; uncorrected tests; no Dawid-Skene.
- **No visual grounding:** real image features (e.g. CLIP, ViT) are the most likely way to recover a content-dependent signal.

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